

Monitoring Techniques for Evaluating Suspected Sleep-Related Breathing Disorders

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Chapter Highlights

- Polysomnography evolved from a research tool into a medical diagnostic procedure largely applied for diagnosing and treating sleep-related breathing disorders.
- Principal components of an evaluation for sleep-related breathing disorders are airflow measurement, respiratory effort assessment, and oxyhemoglobin desaturation recording. Appreciating the relationship between sleep-disordered breathing events and sleep disruption also provides crucial information for patient care.
- Techniques are available to measure lung volume changes, blood pressure changes, and carbon dioxide; however, they are not routinely used for sleep-related breathing disorder evaluation in adults.

OVERVIEW

Soon after their discovery, sleep-related breathing disorders (SRBDs) became clinical polysomnography's dominant focus.¹ Polysomnography began as a laboratory research technique to study sleep, but with a few additional channels, it quickly found a new role as the "gold standard" modality for diagnosing obstructive sleep apnea. Penetration for assessing sleep-disordered breathing has been so complete that undoubtedly the vast majority of sleep studies performed tonight, or on any given night, will serve this function. Nonetheless, home sleep testing increasingly makes inroads as the technique to diagnose sleep-related breathing disorders.² With few exceptions, polysomnography and home testing devices use similar techniques. This chapter presents an overview of recording techniques and their application for evaluating breathing during sleep.

Abnormal breathing during sleep takes several different forms and arises from various underlying etiologies. It goes by many names, including sleep apnea, sleep apnea-hypopnea syndrome, sleep-disordered breathing (SDB), sleep-related breathing disorder (SRBD), periodic breathing, Cheyne-Stokes respiration (CSR), and hypoventilation (Box 164-1). However, obstructive forms of sleep apnea-hypopnea syndrome remain by far the most common SRBD. Detailed descriptions regarding SRBD-associated etiology, morbidity, and treatments appear elsewhere in this book.

The specific pathophysiologic events underlying SRBD include apnea episodes, hypopnea episodes, respiratory effort-related arousals, oxyhemoglobin desaturations, and snore arousals. As their names imply, *apnea* is cessation of breathing and *hypopnea* is shallow breathing. The face validity, that not breathing is undesirable for the health of the organism, seems obvious. However, the association between apnea and morbidity can be, but is not necessarily always, a matter of

cause and effect. Other pathophysiologic processes, especially those related to metabolic and cardiac diseases, can produce downstream effects on breathing.

Polysomnographic and home sleep testing criteria for designating a breathing event as an apneic spell have been consistently defined for decades. Cessation of breathing for 10 seconds or longer constitutes an apnea. Ten seconds was chosen because it approximates missing two breaths. The definition for hypopnea, by contrast, remains controversial. *Hypopnea* essentially means "shallow breath." The difficulty in defining hypopnea stems from several sources. The first involves measurement technique, the second from the fact that hypopneas are not intrinsically pathophysiologic, and finally from decades of usage without any standard.

Polysomnographic and home sleep testing airflow monitoring techniques mostly rely on surrogate and uncalibrated measurements (as detailed further on). Furthermore, airflow signal magnitude from thermistors, thermocouples, capnographs, and nasal pressure transducers correlates poorly with tidal volume. Consequently, operational definitions for hypopnea based on a percentage decrease of flow signal use arbitrary cutoff points (regarding which considerable disagreement continues).

Brief hypopneic intervals routinely occur during wakefulness without causing harm. For example, hypopnea accompanies speaking, and except in extraordinary circumstances, talking does not produce adverse health consequences. During sleep, however, a hypopnea may provoke significant oxyhemoglobin desaturation or a central nervous system (CNS) arousal. Thus the consequence of the hypopnea (not the hypopnea itself) conceivably may be deemed pathophysiologic. Accordingly, prerequisites for designating sleep hypopnea as abnormal involve (1) accurately measuring physiologic consequences, (2) determining at what point these consequences reach significance, and (3) demonstrating their role in morbidity.

Box 164-1 SLEEP-RELATED BREATHING EVENTS, DISORDER CLASSIFICATIONS, AND PARAMETERS

- **Apnea (A):** The complete or near-complete cessation of breathing for 10 seconds or longer in an adult.
- **Oxyhemoglobin desaturation event (ODE):** A decrease in blood oxygen saturation. Usually, the magnitude must be 3% or greater to distinguish it from signal noise.
- **Hypopnea (H):** A reduction in (but not cessation of) ventilation. To be clinically significant, the hypopnea must be associated with an oxyhemoglobin desaturation event or a CNS arousal.
- **Desaturating hypopnea (DH):** A hypopnea with oxyhemoglobin desaturation of 4% or greater (also known as a “Medicare hypopnea”).
- **Respiratory effort–related arousal (RERA):** A CNS arousal terminating obstructive breathing events that do not meet the criteria for apnea or hypopnea.
- **Obstructive apnea or hypopnea episode (OA or OH):** Breathing event caused by upper airway obstruction or reduced upper airway patency.
- **Central apnea or hypopnea episode (CA or CH):** Breathing event caused by absent or reduced respiratory effort (i.e., decreased output to inspiratory muscles from CNS respiratory control centers).
- **Mixed apnea or hypopnea episode (MA or MH):** Breathing event with both obstructive and central features. Many laboratories assign mixed events to the obstructive category.
- **Periodic breathing (PB):** A regularly repeating pattern in which normal or increased ventilation alternates with decreased or absent ventilation.
- **Central sleep apnea syndromes (CSASs):** These syndromes include primary central sleep apnea, Cheyne-Stokes breathing pattern, high-altitude periodic breathing, central sleep apnea due to medical conditions other than Cheyne-Stokes, and central sleep apnea due to drug or substance use/abuse.*
- **Obstructive sleep apnea syndromes (OSASs):** These syndromes include obstructive sleep apnea and obesity-hypoventilation syndrome (see Sleep-Related Hypoventilation Syndromes†). Primary snoring is a normal variant of an obstructive airway process.*

*Opiate and other sedatives are known to depress respiration by blunting respiratory drive.

†ICSD-2 diagnostic classification.

CNS, Central nervous system.

Unfortunately, measurement and morbidity determination issues have remained unresolved for many decades. Many operational definitions for hypopnea* have appeared in published clinical and research literature. Emergence of differing criteria for percentage airflow decrease and oxyhemoglobin decrease and inclusion versus exclusion of CNS arousal as part of the definition seriously complicated matters. Definitional criteria set forth in a standardization attempt sponsored by the American Academy of Sleep Medicine, often referred to as the “Chicago Criteria,” were adopted for research but not for clinical use.³

Subsequently on April 1, 2002, like it or not, the Centers for Medicare and Medicaid Services (CMS) defined a

hypopnea as “an abnormal respiratory event lasting at least 10 s with at least a 30% reduction in thoracoabdominal movement or airflow as compared to baseline, and with at least 4% desaturation.”⁴ Eliminating CNS arousal from the hypopnea criteria and requiring 4% desaturation (rather than 3%) decreased the number of scorable hypopneic spells in many patients.^{5,6}

On the face of it, respiratory-related and snore-related arousals certainly seem irrefutably abnormal (because they compromise sleep integrity); however, issues have arisen concerning detection reliability (and impossibility with use of many home sleep testing devices). Ignoring CNS arousal provoked by respiratory events seemed so ridiculously misguided to many clinicians that they began tabulating such events (even though they fell short of the 4% desaturation criteria). The designation *respiratory effort–related arousals* (RERAs) was commandeered from the Chicago Criteria even though it originally had quite different criteria.

In a major effort to standardize sleep medicine clinical practice, the AASM published *The AASM Manual for the Scoring of Sleep and Associated Events: Rules, Terminology and Technical Specifications* in 2007,⁷ sometimes referred to as the *AASM Manual*. With regard to hypopnea, two definitional criteria were offered, neither of which agreed with the CMS criteria. Member centers were required to adopt and use the AASM criteria to maintain accreditation. One of the definitional criteria used a 3% oxyhemoglobin desaturation and the other included CNS arousal. These conflicting mandates and ambiguities placed American sleep specialists in an intellectual, clinical, and legal dilemma. Clinicians seeing even a single Medicare- or Medicaid-covered patient are required to use CMS criteria for all patients seen in their practice. Failure to comply with this regulation constitutes Medicare fraud. This CMS directive’s intent is to prevent a two-tier medical system from developing. AASM issued clarifications of the rule in 2012, and ultimately, in 2013, AASM adopted CMS criteria⁸ for scoring hypopnea.

Mitterling and colleagues examined scoring outcome using the different definitions.⁹ Applying the 2012 scoring criteria produced higher apnea-plus-hypopnea indices compared with scoring based on the 2007 rules. These investigators used a sample of 100 healthy sleepers, ranging in age from 19 to 77 years.

Apnea and hypopnea episodes can be further categorized as obstructive, central, or mixed, depending on the presence or absence of respiratory effort during the entirety or some part of the breathing event. Beyond mere presence or absence, changes in the effort’s magnitude can provide information concerning airway resistance. Increasing effort leading to a CNS arousal provides insight regarding pathophysiology. Therefore respiratory effort represents a crucial measure for evaluating patients with SRBD.

Sleep-disordered breathing severity can be based on a clinical dimension (e.g., sleepiness), event frequency (e.g., number of events per hour), or magnitude of the consequence (e.g., degree of oxyhemoglobin desaturation). Table 164-1 provides examples for dimensionally classifying severity of sleep-disordered breathing. General agreement is lacking on assignment of severity descriptors to indices of sleep-disordered breathing; however, two schemes are commonly used. In the first, “liberal” classification, an apnea-hypopnea index (AHI) between 5 and 15 is mild, between 15 and 30 is

*At last count, this author (MH) is aware of more than 15 different definitions for hypopnea.

Table 164-1 Clinical/Laboratory Features of Obstructive Sleep Apnea Syndrome by Severity

Dimension	Mild	Moderate	Severe
Sleepiness or unintended sleep episodes	During activities requiring little attention (e.g., watching television)	During activities requiring some attention (e.g., business meeting)	During activities requiring active attention (e.g., driving)
PSG SRBD events: number per night	5–15	15–30	>30
PSG/HST SRBD events and oxyhemoglobin (SaO ₂)	RDI or AHI 2–20 and/or SaO ₂ nadir >85%	RDI or AHI 20–40 and/or SaO ₂ Nadir 65%–85%	RDI or AHI >20 and/or SaO ₂ nadir <65%

AHI, Apnea-hypopnea index; HST, home sleep test; PSG, polysomnography; RDI, respiratory disturbance index; SaO₂, arterial oxygen saturation; SRBD, sleep-related breathing disorder.

moderate, and greater than 30 is severe. In the second, “conservative” classification, an AHI between 10 and 20 is mild, between 20 and 50 is moderate, and greater than 50 is severe.¹⁰

MEASURING AIRFLOW

General Considerations

Most clinical techniques use qualitative surrogate measures to estimate airflow changes. Fully quantitative airflow determination requires pneumotachography or having the patient sleep in a “body box”; however, such techniques are unsuitable for routine clinical sleep studies. Although semiquantitative measures are attainable using calibrated inductance plethysmography, most clinical evaluations rely on qualitative nasal-oral thermography and nasal pressure. This approach provides adequate data and minimizes patient discomfort, reduces costs, and simplifies data acquisition. The AASM Manual recommends using a thermal sensor to identify apnea and a nasal pressure sensor to detect hypopnea.⁷ However, other methods also provide reliable assessment. Airflow also can be measured qualitatively by detecting chemical differences between ambient and expired air (e.g., capnography). However, respiratory activity requires careful classification. Sometimes the patient’s airway can be completely occluded during inspiration but release small puffs upon expiration (detectable by thermistor or CO₂ analyzer). Such events are erroneously categorized as hypopneas or even normal (unobstructed breathing). Figure 164-1 illustrates the problem.

Measuring Temperature

Exhaled air usually is warmer than ambient temperature. Air in the lungs is warmed by core body heat, thereby creating a temperature difference between air entering and exiting the respiratory system. Consequently, measuring temperature fluctuation at the nares and in front of the mouth provides a simple surrogate measure of airflow. Measurement is possible using several different technologies.

Thermistors are thermally sensitive variable resistors that produce voltage alterations when connected in a low-current (but constant-current) circuit. Low current minimizes the tendency for the thermistor to heat itself. Thermistors maximize sensing area while minimizing sensor size and mass. Small temperature changes can produce large resistance changes that can in turn be transduced with a bridge amplifier. Care must be taken to ensure that the thermistor remains below body temperature (i.e., it must not rest on the skin); otherwise, expired air will not be warmed, and no resistance

ERROR IN DETECTING APNEA

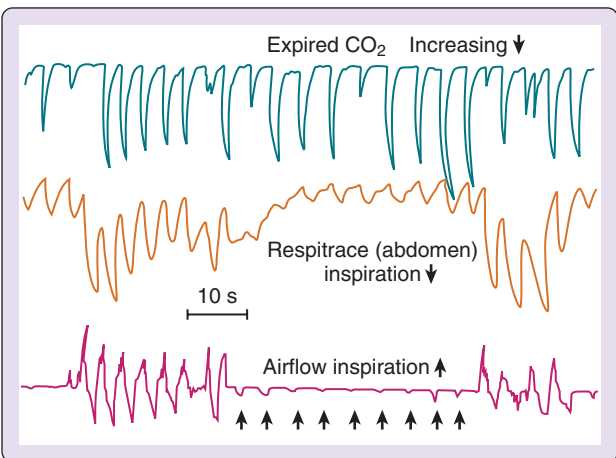


Figure 164-1 An example of the limitations of noninvasive airflow detection. Airflow is recorded simultaneously with a CO₂ analyzer and a pneumotachograph. During the obstructive apnea event, periods of expiratory airflow occur (recorded by the pneumotachograph and the CO₂ analyzer) in the absence of inspiratory flow (obvious in the pneumotachograph recording and unclear in the CO₂ recording). Without the information from the pneumotachograph, the recording from the CO₂ analyzer would be interpreted as evidence of uninterrupted inspiratory and expiratory airflow. *Top*, Airflow is detected with the CO₂ analyzer. *Middle*, Respiratory inductance plethysmograph (RIP) (“Respirace”). *Bottom*, Airflow measured with a pneumotachograph. With each apnea-related expiratory deflection documented by the pneumotachograph (arrows, *bottom*), there is a sustained shift in the baseline of the RIP tracing. This correlation suggests an incremental decrease in functional residual capacity resulting from absence of inspirations with continued small expiratory puffs. If only the top two tracings were available, this pattern would have been mistakenly called hypoventilation or hypopnea, whereas it clearly reflects total occlusion on inspiration. (From West P, Kryger MH. Sleep and respiration: terminology and methodology. *Clin Chest Med* 1985;6:706.)

change will occur. In such a case, inspiratory activity and a respiratory pause will not be differentiable.

Thermocouples also sense temperature change but use a different approach. Different metals expand at different rates when heated. This difference can be transduced to voltage alterations displayable on polygraph systems. Like thermistors, thermocouples are placed in the airflow path in front of the nares and mouth, where expired air heats the sensor and increases its resistance. The transduced signal reflects oscillation between exhaled warm air and cooled inhaled air, thus providing a trace roughly corresponding to respiratory airflow.

Nasal Airway Pressure

During inspiration, airway pressure is negative relative to atmosphere. By contrast, expiration produces a relatively positive pressure in the airway. The resulting alteration in nasal airway pressure can provide a surrogate estimate of airflow and correlates favorably with pneumotachographically recorded signals.¹¹ The nasal pressure signal also offers greater sensitivity than that of nasal-oral thermography for detecting subtle flow limitations (Figure 164-2) when patients breathe through their noses.¹² Airflow limitation shows as pressure trace plateauing during inspiration. A direct current (DC) amplifier provides optimal interface; however, long time-constant alternating current (i.e., a very slowly coupled signal) can suffice. By contrast, rapid coupling can create artifact (Figure 164-3).

Expired Carbon Dioxide Sensors

CO₂ concentration in air leaving the lungs far exceeds that in ambient air. Thus measuring CO₂ in front of the nose and mouth can detect expiration. Infrared analyzers can determine the concentration. Because exhaled CO₂ reflects physiologic chemical change, it offers several advantages compared with physical changes detected by thermistor, thermocouple, and nasal pressure recordings.

In some patients, the end-of-breath CO₂ concentration provides evidence of elevated end-tidal PCO₂. The catheters sampling CO₂ typically entrain some room air, making the measured CO₂ lower than actual end-tidal PCO₂. Therefore an elevated CO₂ indicates that true PCO₂ is even higher, thereby providing a noninvasive technique (merely sampling the air stream) for detecting hypoventilation. The shape of the expired CO₂ curve can also offer useful information. When the patient's baseline expired CO₂ curve shows a clear-cut plateau, the loss of this plateau (or the curve's becoming smaller or dome-shaped) indicates a change in breathing pattern, usually a reduction in expiratory volume.

During central apnea, a low-volume catheter system set at its most rapid response time can show cardiogenic oscillations in the CO₂ signal. These oscillations result from small volume displacements caused by the beating heart.¹³ These heartbeat-synchronized oscillations signify upper airway patency (Figure 164-4).

Capnography can be used in the sleep laboratory to titrate noninvasive ventilation in patients with hypoventilation syndromes.¹⁴ In infants and children with upper airway obstruction, severe hypoventilation may occur during sleep without observable apnea or hypopnea. Measuring expired CO₂ provides evidence for hypoventilation not detectable using thermistors or thermocouples.¹³

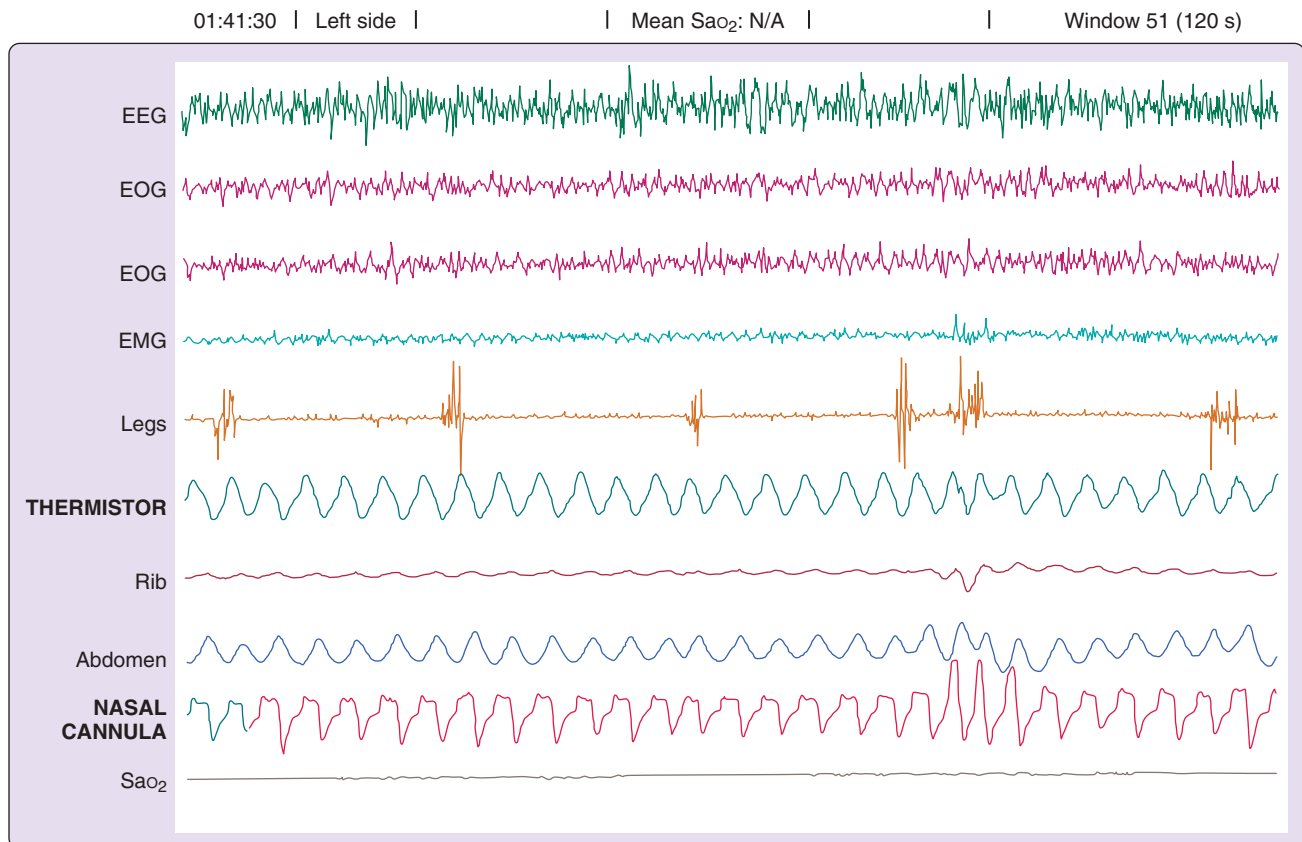


Figure 164-2 A 120-second section from a nocturnal polysomnogram in a subject undergoing simultaneous recording with a conventional thermistor and with a nasal cannula used for recording pressure. Nothing in the thermistor tracing suggests a respiratory event, and subtle movement barely registers in the rib and abdominal inductance plethysmographic tracings. In the nasal cannula tracing, however, the end of one flow limitation episode and the beginning of another are easily detected. Note the plateaus (*chopped-off tops*) of the pressure traces during flow limitation. EEG, Electroencephalogram; EMG, electromyogram; EOG, electrooculogram; SaO₂, oxygen saturation in arterial blood. (Courtesy Dr. David Rappaport, New York University, New York.)

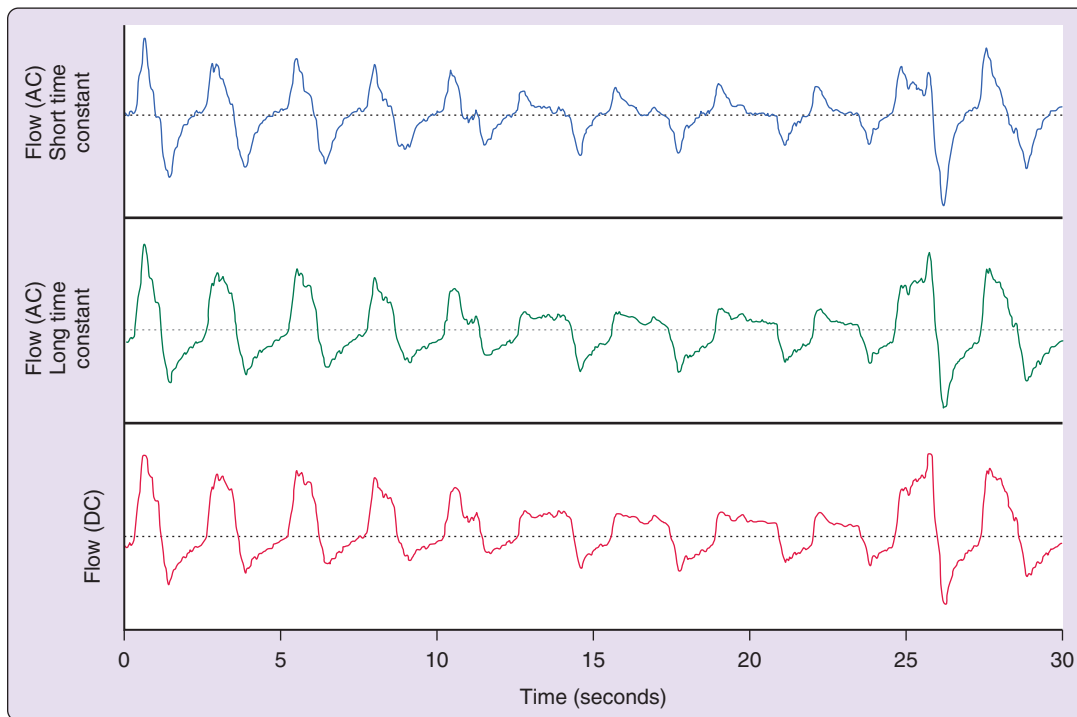


Figure 164-3 A flow limitation event recorded from the nasal cannula measuring pressure simultaneously amplified by three different amplifiers. The bottom signal is from a direct current (DC) amplifier with no filtering. The top two signals are from alternating current (AC) amplifiers with low-frequency filters, with time constants of 1.6 (*top*) and 5.3 (*middle*). The shorter time-constant filter (*top*) causes the flow signal to decay to baseline rapidly during a period of relatively constant flow (flow limitation plateau). The longer time-constant filter (*middle*) provides reasonably good reproduction of these constant flows.

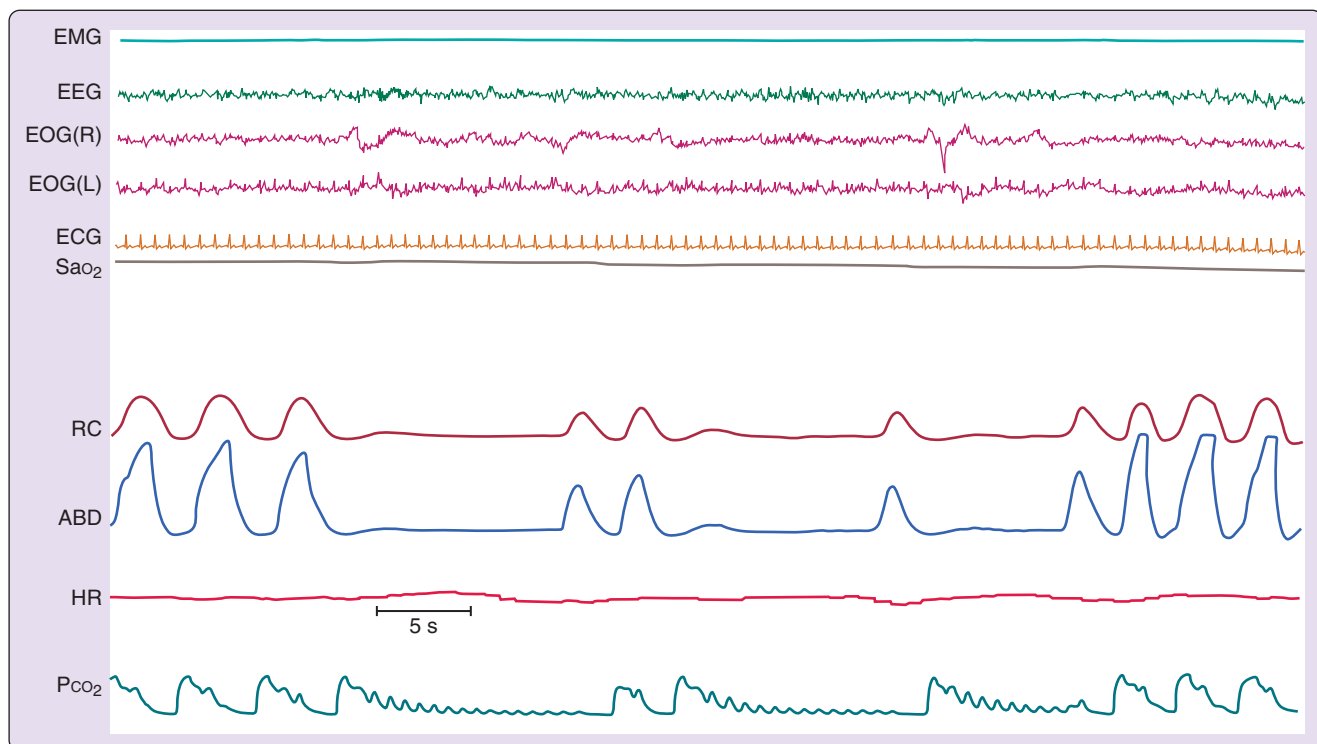


Figure 164-4 Cardiogenic oscillations in CO_2 are seen in the *bottom channel* of the recording in this example of central apnea. The presence of these oscillations, synchronous with the heartbeat, signifies that the upper airway is patent. ABD, Abdominal (movement); EEG, electroencephalogram; ECG, electrocardiogram; EMG, electromyogram; EOG, electrooculogram; HR, heart rate; RC, rib cage (movement); PCO_2 , partial pressure of carbon dioxide; SaO_2 , oxygen saturation in arterial blood.

Pneumotachography

Pneumotachography accurately and quantitatively measures airflow volume. The patient usually wears a face mask, and the procedure can be uncomfortable. In awake subjects, pneumotachography is known to alter breathing; it increases tidal volume and reduces respiratory rate. Some positive airway pressure machines contain an integrated pneumotachograph usable to monitor airflow during laboratory titration. Pneumotachography is seldom used for routine SRBD diagnostics.

Several types of pneumotachographs are available. These devices differ with respect to measurement technique: They use either (1) differential pressure airflow transducers, (2) ultrasonic flow meters, or (3) hot-wire anemometers. Discussion is limited here to differential pressure flow transducers because they are the most widely used. In this technique, airflow directed through a cylinder exits through a small resistive field, usually composed of small parallel tubes or a grill promoting laminar flow. The pressure drop across this resistive field is measured using a differential manometer. When flow is laminar, the relationship between the pressure differences and flow is linear. Changes in gas density, viscosity, and temperature alter the pressure-flow relationship. To prevent condensation on the resistive element requires heating, so calibration should be conducted when the pneumotachograph is heated. After correction for errors introduced by alterations in these physical factors, the flow signal is integrated to determine volume.

MEASURING RESPIRATORY EFFORT

Measuring respiratory effort provides information needed to distinguish between respiratory events of obstructive versus nonobstructive (central) etiology. Categorizing an apnea or hypopnea as obstructive, mixed, or central in origin derives from differential respiratory effort and airflow patterns. As reviewed next, several techniques are available to detect and/or measure respiratory effort, including rib cage and abdominal motion, electromyography, pleural pressure changes, movement detected by static charge sensors in or on the bed surface, movement detected by standing wave patterns in the bedroom, and digital video recording.

Rib Cage and Abdominal Motion

Currently, the most common polysomnographic technique for measuring respiratory effort involves detection and quantification of rib cage and abdominal movements. During normal breathing, the major inspiratory muscles produce rib cage expansion and a downward movement of the diaphragm. These movements cause the pressure around and in the lung to become negative (relative to atmospheric pressure). The pressure gradient between ambient air and the lung draws air through the airways into the alveoli. Thus a change in lung volume is the sum of the volume changes of the structures surrounding the lungs, the rib cage, and the abdomen.¹⁵ Other respiratory muscles (e.g., intercostal, sternocleidomastoid) also play a role in stabilizing the thoracic cage. Some clinicians erroneously interpret the abdominal and rib cage motion changes as implying separate activities of abdominal and thoracic respiratory muscles, but this is not the case. Virtually all of the changes in abdominal and rib cage volumes (including paradoxical motion) can be explained by changes in the status of the respiratory muscles directly inserting onto the thoracic

Box 164-2 MECHANISMS UNDERLYING PARADOXICAL MOTION OF THE RIB CAGE AND ABDOMEN

Loss of diaphragm tone. When the diaphragm ceases to contract and becomes flaccid, it merely reacts to pressure changes around it instead of generating pressure changes. In this situation, when the other respiratory muscles contract, the rib cage is enlarged, and pleural pressure becomes negative, sucking the diaphragm into the chest. This condition results in an increase in rib cage volume and a reduction in abdominal volume.

Loss of accessory respiratory muscle tone. When the accessory muscles lose tone, the rib cage, particularly the upper part of the rib cage, becomes unstable. When the diaphragm then contracts, the negative intrathoracic pressure causes the unstable part of the thorax to be sucked in during inspiration.

Partial upper airway obstruction. With partial upper airway obstruction, the diaphragm must generate very strong negative pressures for inspiration to occur. As the diaphragm contracts, it both pushes out the abdomen and creates great negative intrathoracic pressure. This highly negative intrathoracic pressure can overcome the mechanisms maintaining chest wall stability (accessory muscle tone and rigidity of the cage), so that the least stable portions of the rib cage will tend to move inward with inspiration. This potential for inward movement of the rib cage is a problem mainly in the very young, in whom the rib cage is quite pliable.

cage. Paradoxical motion of the rib cage and abdomen can result from several changes, including loss of tone of the diaphragm, loss of tone of the other respiratory muscles, and upper airway obstruction (complete or partial). The mechanisms underlying this asynchronous motion of rib cage and abdomen are described in Box 164-2. Regardless of the pattern or its underlying mechanism, rib cage and abdominal movement reflect effort to breathe.

At a minimum, a single uncalibrated abdominal movement sensor can *detect* respiratory effort. Respiratory effort during airflow cessation usually signifies airway obstruction. Common approaches for *measuring* rib cage and abdominal movement use (1) strain gauges, (2) inductance plethysmography, and (3) piezoelectric transducers.

Strain gauges are sealed elastic tubes filled with conductive material through which an electric current is passed. When length is constant, current and resistance are constant. Stretching the strain gauge lengthens and narrows the cross-sectional area of the fixed-volume conductor. This deformation produces a proportional increase in electrical resistance. Current varies inversely in relation to the length of the gauge, thereby becoming an index of gauge length. A Wheatstone bridge amplifier transduces this change to voltage for continuous display showing rib cage or abdominal expansion (depending on placement).

Inductance plethysmography electronically measures changes in the cross-sectional area of the rib cage and abdominal compartments by determining changes in inductance. Inductance is a property of electrical conductors characterized by the opposition to a change of current flow in the conductor.

Transducers are placed around the rib cage and abdomen—the physiologic equivalent of conductors. Each transducer consists of an insulated wire sewn into the shape of a horizontally oriented sinusoid and onto an elasticized band.

Piezoelectric transducers are used in yet another method of detecting movement. These sensors can be placed on the rib cage and abdomen and are sensitive to changes in length. When a piezoelectric crystal is squeezed, an electrical potential appears across its sides. The crystals can be arranged (usually as part of a belt) so that movement can be detected (see next).

Combined Sensors

Thermocouples, thermistors, and strain gauges are fairly old technologies. More recently, new materials such as polyvinylidene fluoride film have been introduced into the sleep laboratory. Such film has the interesting property of converting heat and mechanical energy into electrical energy that can be measured. Polyvinylidene fluoride films have both piezoelectric (responding to mechanical changes) and pyroelectric (responding to thermal changes) properties. The output from these films can be configured to measure airflow¹⁶ (pressure and temperature), snoring¹⁷ (pressure waveforms), and changes in length caused by abdomen and rib cage movement.¹⁸⁻²⁰

Respiratory Muscle Electromyography

Recording intercostal muscle activity on the electromyogram (EMG) is one of the oldest polysomnographic techniques for detecting respiratory effort (Figure 164-5). These uncalibrated recordings are made using standard surface electrodes placed in pairs in the intercostal spaces on the right anterior chest. Obtaining an optimal signal requires practice, patience, and skill; recordings are prone to artifact, especially artifact from the electrocardiogram (ECG). Intercostal EMG activity, when recorded properly, can be extremely valuable for differentiating among central, obstructive, and mixed sleep-disordered breathing events. Furthermore, although signals are not calibrated, cascading increases in respiratory effort are readily apparent from recordings.

Pleural Pressure Changes

Some sleep centers use esophageal pressure to index inspiratory effort. In our experience, most patients undergoing all-night polysomnography find esophageal balloons unacceptable. However, the thin water-tip or catheter-tip piezoelectric transducers are better tolerated. Esophageal pressure measurements help verify central apnea or hypopnea episodes with a high degree of certainty. This technique also can detect very subtle respiratory events. However in subtle cases, a repeat

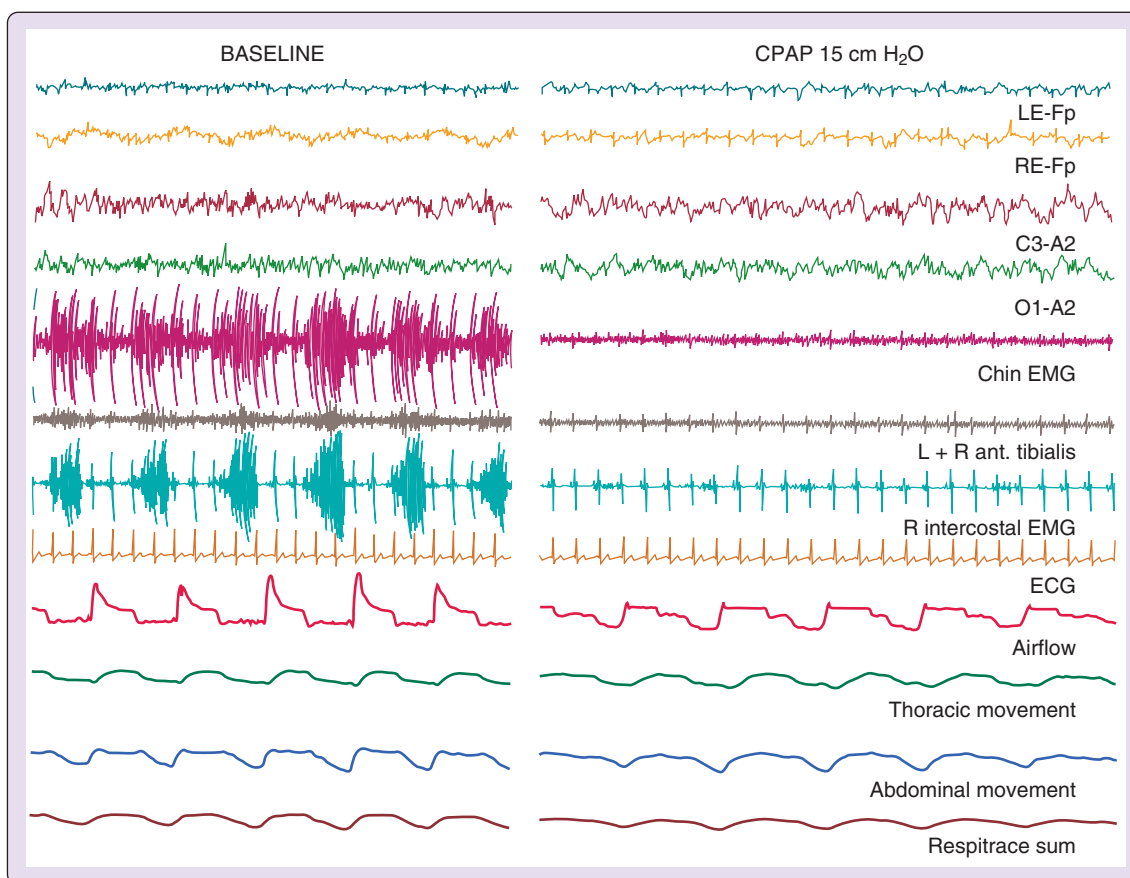


Figure 164-5 Surface respiratory (R intercostal) muscle EMG in sleep apnea. *Left*, The respiratory EMG signal is dramatically increased. *Right*, This signal is reduced on nasal continuous positive airway pressure (CPAP). A2, Right mastoid reference; C3, left central EEG; ECG, electrocardiogram; EEG, electroencephalogram; EMG, electromyogram; LE-fp, left eye referenced to frontal pole; L + R, linked left and right; O1, right occipital EEG; RE-fp, right eye referenced to frontal pole. (Courtesy Dr. J. Catesby Ware, Eastern Virginia Medical School, Norfolk, Virginia.)

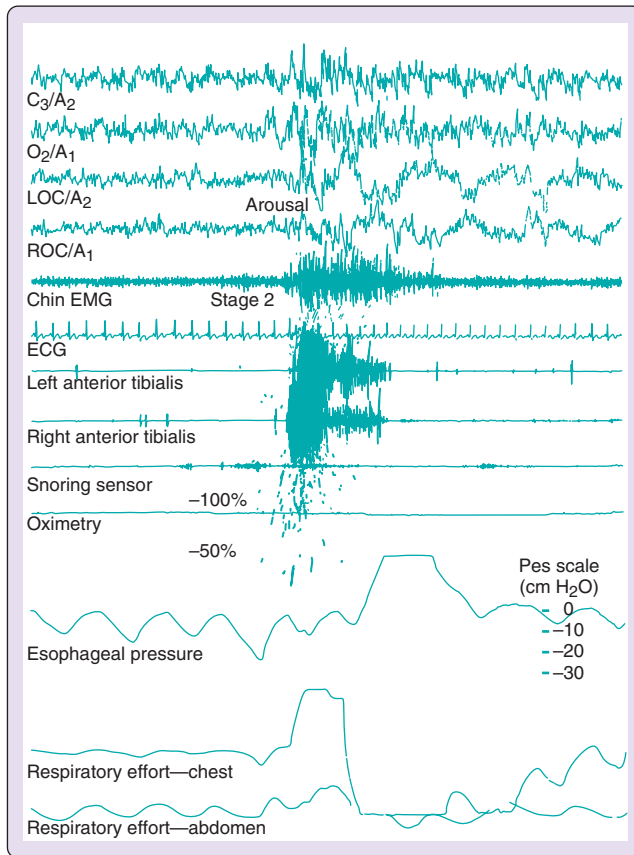


Figure 164-6 Esophageal pressure in the upper airway resistance syndrome. The esophageal pressure (Pes) swing was greatest just before the arousal. C3/A2, Left central EEG referenced to right mastoid; ECG, electrocardiogram; EMG, electromyogram; LOC, left outer canthus; O₂/A₁, right occipital lobe EEG referenced to left mastoid; ROC, right outer canthus. (From Butkov N. *Atlas of clinical polysomnography*. Ashland [Ore.]: Synapse Media; 1996, p. 224.)

polysomnogram may be preferable to using an esophageal catheter.²¹

In patients with *upper airway resistance syndrome*, the classic findings include progressively more negative pleural pressure until occurrence of a CNS arousal. The arousal is sometimes associated with an audible snort (Figure 164-6). After arousal, pleural pressure swings temporarily decrease until the next cycle begins, upon which pressure swings increase until another arousal occurs. Sometimes the CNS activation falls short of AASM duration criteria for arousal. However, the AASM's 3-second rule was established for determining reliability of spontaneous arousals using visual scoring.^{22,23} On the electroencephalogram (EEG), signal changes of shorter duration are conceptually thought to represent CNS arousals, as are spectral analysis indices of alpha power loading, even when it is difficult to see on the raw data tracing.

Movement Detected by Static Charge and Pressure Sensors

The static charge-sensitive bed technology has been evaluated in sleep disorders.²⁴ The transducer is embedded within a thin mattress that responds to the slightest movement. Output from the bed is sensitive enough to detect heartbeat as a ballistocardiogram. Respiratory signal amplitude differs with body position changes; however, output is otherwise stable.



Figure 164-7 Synchronized digital video can be extremely helpful, as in this example in which a child with retrognathia slept with his neck arched and his mandible thrust forward (arrow). This sleep posture resulted in an unoccluded upper airway. The conventional polysomnography recording missed that a significant sleep breathing problem was present. (From Banno K, Kryger MH. Use of polysomnography with synchronized digital video recording to diagnose pediatric sleep breathing disorders. *CMAJ* 2005;173:28–30.)

A similar but newer technology uses piezoelectric sensors embedded in a strip placed perpendicular to the sleeper's body. Changes in pressure resulting from breathing movement and heartbeat can be analyzed.²⁵

Movement Detected by Wave Technologies

Several systems approach movement detection using microwaves, radar, and/or changes in standing wave patterns in the bedroom. One approach directs a beam at the bed surface and analyzes the returning signal to evaluate the sleeper's upper body movement. In one such system, laser radiation can be used. Microwave and other radar-like technologies also can be applied. Additionally, a series of sensors could be used to monitor movement. Application of some of these emerging technologies to sleep medicine for evaluating SRBD is an important aspect of ongoing developments in the field.

Polysomnography-Synchronized Digital Video

Digital video is now a common feature of computerized polysomnography. Although recordings are widely used to evaluate parasomnias and seizures, they can be helpful in assessing SRBD. When the recording is properly synchronized with polysomnography, ambiguous and difficult-to-interpret tracings often become obvious. For example, it is easy to recognize a small dip in arterial oxygen saturation (Sao₂) as a significant sleep-disordered breathing event when it is followed by oxygen resaturation after an audible snort, a repetitive moving forward of the jaw, an arching of the neck, or a closing of a gaping mouth. Video is especially helpful in children, whose polysomnographic recordings may be difficult to interpret (Figure 164-7). An Sao₂ dip, without the other visual information, is quite likely to be missed, ignored, or dismissed as artifact. Video recordings are particularly useful in thin persons, who might not experience oxygen desaturation with their abnormal respiratory events. Furthermore, showing the sleep study video to patients can be very effective for promoting understanding of the problem and an appreciation of its severity.

MEASURING CHANGES IN LUNG VOLUME

Several methods can estimate tidal volume. These measures provide semiquantitative data concerning presumed airflow while also documenting respiratory effort. Calibration is

crucial in attempting to use these devices to gauge ventilation. Postcalibration movements, changes in body position, and shifting in placement of the recording device can introduce error. In sleep laboratories, strain gauges, inductance plethysmography, and impedance pneumography are sometimes used to measure volume changes. Other techniques include magnetometry, body plethysmography, canopy with neck seal, the barometric method, and pneumotachography; however, these are seldom used in a clinical setting.

Strain Gauges, Inductance Plethysmography, and Piezoelectric Transducers

In principle, length-sensitive devices can be used qualitatively to detect breathing abnormalities. If properly calibrated, these devices can be used quantitatively to measure dynamic volume changes.²⁶ Normally, the enlargement of the thorax and the outward movement of the abdominal wall occur together; that is, they are in phase. For a given change in lung volume, then, it is possible to quantify a change in rib cage volume and abdominal volume. For a given breath, the relative contributions of the rib cage and abdominal compartments also can be determined.

To quantify actual volume changes, the transducers must be calibrated against an independent volume-measuring system. In practice, two length-measuring devices are required for measuring changes in lung volume: one for the rib cage and one for the abdomen. The rib cage device is placed at the level of the axilla, and the abdominal device is placed just superior to the iliac crest. If it is assumed that the fractional contributions of the abdomen and the rib cage are constant, then changes in lung volume can be measured by calibrating transducers sensitive to rib cage and abdominal displacement. Once the transducers are calibrated, the sum of the rib cage and abdominal excursions will describe volume changes.

Unfortunately, the relative rib cage and abdominal contributions can change with posture and muscle tone occurring in sleep. Movement-related device migration from its original site and device deformability also must be considered. Such factors adversely affect calibration accuracy and stability. Nonetheless, calibrated inductance plethysmography appears to be sensitive enough to detect upper airway resistance syndrome events.²⁷

Impedance Pneumography

Impedance defines the combined effects of two previously discussed properties of an electrical conductor: *resistance* and *inductance*. In physical terms, when impedance pneumography is used, the conductor is the thorax. Impedance is measured by applying a small current across the thorax using a pair of electrodes placed at the site of maximal thoracic excursion.

Transthoracic impedance changes reflect variations in the amount of conductive materials (liquids, including interstitial fluid, blood and lymph, and tissue) and nonconductive material (air) between the electrodes. The conductive and nonconductive materials affect the total impedance differently. Increased air in the lung increases impedance, and increased fluid in the thorax decreases impedance. A recording of the volume of air exchanged and the total of impedance changes can allow the differentiation between air-related and fluid-related changes in impedance. If total impedance is recorded in a single channel, changes related to air volume and fluid are measured.

Impedance alterations in obstructive apnea are complex. During apnea, lung volume decreases, whereas the negative intrathoracic pressure most likely temporarily pools blood in the pulmonary circulation. For these reasons, a precise measurement of respiratory volume and pattern may not be possible. Nonetheless, rate-adapting cardiac pacemakers using transthoracic impedance to drive ventilation have been used to screen for obstructive sleep apnea.²⁸

MEASURING THE PHYSIOLOGIC CONSEQUENCES OF SLEEP-RELATED BREATHING DISORDERS

As previously mentioned, some SRBD events (e.g., hypopnea) are not intrinsically pathophysiologic; however, their consequences are. SRBD event consequences include oxyhemoglobin desaturations, carbon dioxide elevations, blood pressure changes, electrocardiographic abnormalities, and CNS arousals.

Oxygen and Carbon Dioxide Alterations

Routine clinical sleep evaluations require noninvasive, continuous, rapidly sampled measures to determine blood oxygen concentration. Thus directly measuring blood oxygen with an indwelling arterial catheter is ruled out on all counts. Pulse oximetry, however, serves the need well; consequently, it has become the standard technique for recording oxyhemoglobin desaturations during sleep. Pulse oximeters usually determine Sao_2 spectrophotoelectrically using a two-wavelength light transmitter and a receiver placed on either side of a pulsating arterial vascular bed (usually a finger, a toe, an ear, or the nose). Alternatively, in reflectance pulse oximetry, the light transmitter and the receiver are on the same surface. The light transmitted into the vascular bed is scattered, absorbed, and reflected. The amplitude of light detected in particular spectra by the receiver depends on the magnitude of the change in arterial pulse, the wavelengths transmitted through the arterial vascular bed, and the oxygen saturation of arterial hemoglobin (deoxygenated blood is bluer). These devices are sensitive only to pulsating tissues; consequently, venous blood, connective tissue, skin pigment, and bone theoretically do not hamper determination of Sao_2 . Accurate measurement, however, requires a minimal pulse amplitude. Dyshemoglobinemias can cause problems.

Correct alignment of the light transmitter and receiver is critical to ensure measurement accuracy. If the sensor is applied to a digit, that digit must be immobilized. Significant bending of the digit can compromise detection of pulsatile flow and invalidate results. Although all pulse oximeters are based on similar technology, response characteristics differ both across manufacturers and even within a specific manufacturer's product line.²⁹ Sensor placement and device programming are crucial technical factors in obtaining optimal results. Reflectance oximeters also must contend with weaker pulse signals, because much less light is reflected back to the sensor. Differences in response characteristics and effect of sensor location cannot be overemphasized (Figure 164-8), because some oximeters appear to be completely insensitive to hypoxemia episodes clearly detected by other devices (Figure 164-9). Such devices can produce false-negative SRBD test outcomes.

A review of specific oximeters would be beyond the scope of this chapter; nonetheless, generalizations about (1) sensor location, (2) instrument filtering and sampling rates, and (3) potential pitfalls are worth noting here.

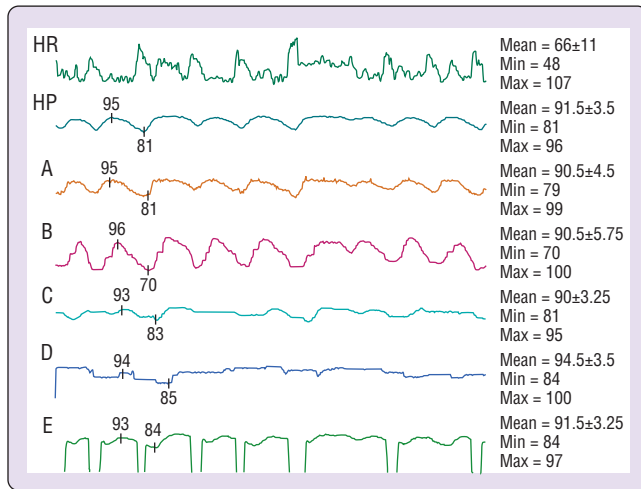


Figure 164-8 Heart rate (HR) and arterial oxygen saturation (SaO_2) during a 5-minute period in a patient with sleep apnea. The top two traces are for the HR monitor and a Hewlett Packard (HP) oximeter; A to E are traces for five different pulse oximeters. The scales for the six oximeters are identical. The numbers on the tracings represent the instantaneous SaO_2 measured during the peak and trough of an apneic episode. The data listed to the right of the figure are the mean, standard deviation, and minimum and maximum values for HR and SaO_2 for the six oximeters. Note that oximeters C and D do not track SaO_2 , and that the recording for oximeter E has numerous artifacts.

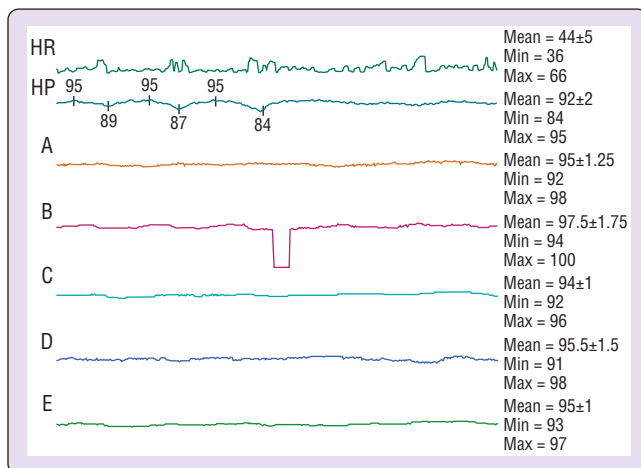


Figure 164-9 Heart rate (HR) and arterial oxygen saturation (SaO_2) during a 5-minute period in a patient with sleep apnea and bradycardia. Tracings are taken from six oximeters, as described in Figure 164-8. In this example, three apneic episodes are missed entirely by all of the pulse oximeters. This patient's problem would have gone completely undetected by pulse oximetry screening. HP, Hewlett Packard (oximeter).

Sensor Location

In our experience, the preferred sensor location in adults is the earlobe. Poor perfusion can be enhanced by applying a trace amount of vasodilator (e.g., nonlytic acid vanillylamide plus nicotinic acid—Finalgon ointment [Boehringer Ingelheim, Ridgefield, Conn.]). Technicians must take care to avoid contact of the perfusion-enhancing agent with their own or the subject's eyes, because it is a powerful cutaneous vasodilator. When the ear site is not usable, reflectance pulse oximeter sensors placed on the forehead or another well-perfused surface will suffice. Recording from the ear also reduces circulator delay (compared with the finger). This advantage

becomes especially important for associating a respiratory event with its subsequent oxyhemoglobin desaturation in patients with congestive heart failure.

Instrument Filtering and Sampling Rate

Most pulse oximeters filter the signal, and some filter algorithms use the heart rate. The degree of filtering is inversely related to heart rate; brief, mild hypoxemic episodes may therefore be missed during periods of very low heart rate, because they are filtered more strongly. During polysomnographic recording, filtering should be minimized (i.e., by setting the oximeter to the fastest response or the highest sampling rate, or both) to reduce missing transient oxyhemoglobin desaturation events.

Potential Problems

Because pulse oximeters use two wavelengths of light to estimate SaO_2 , they cannot distinguish three or more hemoglobin species. In the presence of carboxyhemoglobin (as in heavy smokers, in whom carboxyhemoglobin levels can reach 10% to 20%), SaO_2 is overestimated.³⁰ In the presence of a rising methemoglobin concentration, oximetry-determined SaO_2 will plateau at approximately 85%, regardless of whether true saturation is much higher or lower.³¹ Because light is transmitted through tissue, pigment in the skin can degrade oximeter performance, producing incorrect “probe-off” or “perfusion-low” error reports.³² Some finger-clip devices can produce pressure-related injuries when worn for an entire night.

Partial pressure of arterial oxygen (PaO_2) estimated from the skin's surface depends on oxygen flux through the skin, local oxygen consumption, and the skin's diffusion barrier.³³ This measurement technique is most commonly used in neonates, whose skin is thin. Accurate measurement of transcutaneous Po_2 (Po_{2tc}) requires maximal dilation of the local vasculature in the upper dermis. This is achieved by heating (to 43°C); however, heating shifts the oxyhemoglobin dissociation curve to the right, increases the resistance of the skin stratum corneum to oxygen permeation, increases the metabolic rate of the dermal tissue, and increases the rate of cutaneous blood flow. The shift in the oxyhemoglobin dissociation curve and the increase in metabolic rate effectively cancel each other out, leaving permeability and flow as the dependent factors in correlating Po_{2tc} and PaO_2 . An important advantage of heating is that the amount of blood present is maximal, and Po_{2tc} is therefore unaffected by small changes in blood supply to the tissue.

The Po_{2tc} may be misinterpreted when the state of blood flow is unknown. When flow and PaO_2 are adequate, Po_{2tc} reflects PaO_2 . Under conditions of compromised flow and adequate PaO_2 , Po_{2tc} will change with flow. If SaO_2 and flow are compromised, Po_{2tc} tracks oxygen delivery. Transcutaneous measurement accuracy also depends on correct sensor application. To convert Po_{2tc} measurements to PaO_2 values precisely, a calibration curve for each subject is required, so this technique is too labor-intensive for use in routine clinical practice. In most laboratories, Po_{2tc} serves to track, in relative terms, arterial oxygenation status. Device responsiveness is too slow for tracking rapid blood gas changes associated with brief SRBD events (less than 30 seconds), because oxygen diffuses slowly across the skin. The conditions governing transcutaneous Pco_2 measurement are remarkably similar to those described for PaO_2 . Although transcutaneous blood gas determinations are

of greatest value in neonates and young children, $\text{Pco}_{2\text{tc}}$ is useful for assessing hypoventilation in adults.

Central Nervous System Arousals and Awakenings

Awakenings and CNS arousals provide critical information concerning sleep disturbance and fragmentations. Standardized scoring derives from electroencephalographic activity, preferably recorded from occipital sites. CNS arousals often take the form of bursts of alpha (7 to 13 Hz) activity, and the term *EEG speeding* is sometimes used to describe the event. When scored visually, a 3- to 15-second burst of alpha activity in non-rapid eye movement (NREM) sleep is considered an arousal. During rapid eye movement (REM) sleep, when alpha bursts can be an ongoing part of background activity, the alpha intrusion must be accompanied by increased muscle tone. In any sleep stage, a burst exceeding 15 seconds in duration is scored as an awakening.

The CNS arousal represents a clinically significant sleep parameter because it is associated with (in cross-sectional studies) and provokes (in intervention studies) tiredness, fatigue, and sleepiness. As a pathophysiologic consequence of obstructive sleep apnea, the arousal is thought to result from increased respiratory effort (and accompanying increase in autonomic nervous system sympathetic activation) triggered by increased airway resistance. Presumably, the CNS arousal terminates the SRBD event, because the arousal returns ventilation to voluntary control. As a result, the airway can be dilated and breathing resumes. However, some controversy exists about resumption of breathing after an obstructive event without CNS arousal and about the presence of arousals at the termination of central events.

Blood Pressure Changes

Clinical laboratories do not routinely record sleep-related blood pressure. Nonetheless, many pulse oximeters also can track pulse pressure (a magnitude index of pulsation occurring at the oximetry sensor site). In addition, pulse transit times (PTTs) have been used to indirectly estimate blood pressure.³⁴ PPT represents the time interval from heartbeat (ECG R wave) to its pulse recorded in the periphery. Negative pleural pressure provokes a blood pressure drop and in so doing lengthens PTT. The progressive increase in pleural pressure during obstructive apnea correlates with rising oscillations in PTT amplitude. During central apnea, this does not occur. Thus it has been suggested that PTT might provide an opportunistic estimate of inspiratory effort and thereby provide a means to differentiate obstructive from central apnea.³⁵

Automatic self-inflating arm-cuff sphygmomanometers have long been available but are problematic for routine clinical use because they disturb sleep. By contrast, miniature finger cuff systems disturb sleep much less. Devices obtaining data in alternating fashion from adjacent fingers reduce finger injury risk, and some systems automatically perform hydrostatic correction when arm movement occurs. Finger flexion artifacts, however, remain a problem.

Cardiopulmonary Coupling

Cardiopulmonary coupling uses a single-channel ECG recording to extract signal features modulated by breathing.³⁶ Respiration induces small alterations in heart rate and amplitude variations in R-wave amplitude. Using computerized frequency domain analysis, a high-frequency component can

be isolated that represents variations in breathing-induced vagal sinus pressure heart rate. A low-frequency component can be extracted that provides data about interbreath intervals. By examining a weighted composite of these data (increased variability and the envelope of amplitude change), time-domain periods (usually 2 to 10 minutes) with breathing pauses can be discerned by examining coherence and cross-power spectrums. In general, normal breathing loads high-frequency bands, and SRBD events mass in low-frequency troughs. For detailed information about cardiopulmonary coupling, see Chapter 166.

HOME SLEEP TESTING

In addition to standard attended polysomnography, guidelines for home sleep testing have been published.³⁷ Chapter 165 provides methodologic details. Because home sleep testing devices are less sensitive than laboratory polysomnography, the home test can rule in, but not rule out, SRBD.

To succeed diagnostically and economically, home sleep testing requires proper patient selection, appropriate portable recorder application, study interpretation by a qualified sleep specialist, readily available access to laboratory polysomnography when needed (as after a negative result on home sleep testing, or to investigate continuing problems despite treatment), and systematic follow-up assessments. Many health care coverage plans (including those administered by the Centers for Medicare and Medicaid Services) reimburse home sleep testing for diagnosing sleep apnea. Unattended studies are more prone to data loss as a consequence of uncorrected technical failures (e.g., electrode detachment) and patient tampering. Consequently, only patients with a high clinical suspicion for SRBD should be referred for HST.

CLINICAL PEARLS

- Detecting sleep-related breathing events and classifying each as obstructive, central, or mixed require measures of airflow, respiratory effort, and oxyhemoglobin saturation.
- Information about sleep-disturbing effects of SRBD events provides better sensitivity for the diagnosis of sleep-disordered breathing.
- Standardized guidelines developed by the AASM were developed to advance clinical practice and should not be construed as an impediment to advancing scientific inquiry.

SUMMARY

Evaluating patients with SRBDs is the most common application of laboratory and home sleep studies. Over the past four decades, some of the more frequently used procedures have evolved and become commonplace in clinical sleep laboratories. The American Academy of Sleep Medicine also published guidelines and a standards manual. Standard assessment for breathing disorders during sleep includes measures of airflow, respiratory effort, and oxyhemoglobin desaturation. Laboratory evaluation also includes sleep disturbance assessment. In this chapter we describe techniques to assess these and other, related physiologic activities. The underlying mechanism, advantages, and problems are also discussed.

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A complete reference list can be found online at ExpertConsult.com.

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REVIEW QUESTIONS

1. All of the following are in standard use for evaluating sleep-related breathing disorders *except*:
 - A. Pulse oximeter
 - B. Nasal pressure sensor
 - C. Capnograph
 - D. Respiratory effort bands
2. Which of the following consequences of sleep-related breathing disorders is/are determined on laboratory evaluations but not usually on home sleep tests?
 - A. Autonomic nervous system activation
 - B. Central nervous system arousals
 - C. Nasal pressure changes
 - D. Blood pressure changes
3. A laboratory technique that can be used to calculate an estimate for lung volume is:
 - A. Impedance pneumography
 - B. Cardiopulmonary coupling
 - C. Static charge-sensitive bed
 - D. Microwave radar
4. Which of the following statements about home sleep testing is correct?
 - A. Home sleep testing is more sensitive than laboratory polysomnography.
 - B. Home testing can rule out sleep-related breathing disorders.
 - C. With negative results on home sleep testing, follow-up polysomnography is indicated.
 - D. Home sleep tests usually record airflow, respiratory effort, oximetry, and central nervous system arousals.

ANSWERS

1. C.
2. B.
3. A.
4. C.